

Giving oral presentations is a skill that you can learn. It gets better with practice. Of course, you will be nervous at first, but this can be minimized with planning and practice. I have assembled below some guidelines I have developed for my laboratory over the years that will help to achieve an effective and engaging oral presentation. Note that there are lots of different types of oral presentations. This is a style that works for a presentation in which one is presenting data to a scientific audience, such as your lab, your department or at a professional meeting.

1. Big Picture Planning

1.1. Know your audience

- Present information as you want the audience to see it.

1.2. Tell a story.

- Do not give a book report. A story has a narrative, characters, challenges, questions, drama. A book report is a list of experiments.

1.3. State your Big Biological Question (BBQ) early.

- Use this to motivate your story.

1.4. Establish a flow

- Follow a logical progression. Successful flow has a compelling story line and a consistent color scheme for images.

2. Timing

2.1. For a 10-minute time slot

- This time slot is like a Platform talk at the BPS and like the one you will give in May). You have **8-9** minutes for the actual presentation and **1-2** minutes for the question-and-answer period.
- Prepare 9-12 slides, no more.
- Two big picture background slides
- One "Questions" slide
- Six data slides
- One conclusions slide

2.2. For a 15-minute time slot

- This time slot is like a Platform talk at the BPS and like the one you will give in May). You have **11-12** minutes for the actual presentation and **3-5** minutes for the question-and-answer period.
- Prepare 12-15 slides, no more.
- Two to three big picture background slides
- One "Questions" slide
- Six data slides
- One conclusions slide

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2.3. For a 30-minute time slot (like the thesis talks you will give to the department next year), you are expected to use **25** minutes for the actual presentation (no more) and **5** minutes for the question & answer period.

- Prepare no more than 25 slides.
- 4-5 slides for Background
- One “Questions” slide that lists 3-4 questions you will address. (three is ideal).
- 4-5 slides to address each question
- One general conclusions/future directions slide that brings everything together
- One Acknowledgements slide
- Estimate about one minute per slide. You will be surprised how long that is.
- PowerPoint lets you practice your timings. You must practice.

3. General Slide Guidelines

3.1. Slides should generally have no words.

- With a few slide-type exceptions, slide should have No Words except for the slide title. The *reason* for this rule is that the audience cannot listen to what you are saying and read the words on your slide at the same time. Some people will be reading and some people will be listening but very few people can simultaneously do both. If you have words in the content part of your slide, you must read them so that there is no discordance in the visual and auditory input to the audience. Generally, this is boring, so do not do this.
- Instead of words, you should convey your ideas and data using graphics.

3.2. Background Slides

- Background slides provide context to set up the important questions you will address in your dissertation work.
- These start broad and then become more specific as you approach your BBQ slide.

3.3. Structure of the BBQ slide.

- This slide states the Big Biological Questions of your work.
- This slide is basically the outline for your talk. It serves the purpose of establishing a flow and overall structure for your talk, which sets up the expectations for your audience.
- A question slide is where you lay out what the individual questions you want to address in your talk are.
- Words are allowed on this slide. And, you must read these words.
- This slide should come after your introduction.
- All questions should have been at least partly motivated by your introduction.
- You will use this slide more than once. When you answer the first question, come back to the BBQ slide and put an answer to the question. Then grey this font out and move to the text questions. This presentation technique gives you a transition to the new section.

3.4. Use Informative Slide Titles

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- Each slide should have an informative title summarizing what you want the audience member to take away from the slide
- A title is short, declarative sentence that summarize the slide. Even if the viewer understands nothing about the rest of your slide, they should come away with your title “sound bite”.
- This means that you need to think about what one, single point you want each slide to convey. You condense this point down into only one sentence.
- The short declarative sentence should fill in the *blah blah blah* in the following sentence: “What I want you to know is *blah blah blah blah blah*.”
- If you are struggling with a slide and can’t get it down to one point in a short declarative sentence, step back and rethink the point of that slide and figure out how to distribute the content of the slide in a way that you can summarize each slide in one sentence. Sometimes this means you break your slide into two slides.
- Your short declarative sentence should ideally fit on one line. Two lines is OK. Avoid three lines.
- The title of a slide should be font size 28.
- Never go below 24 for your title. If you have to go to a lower font to fit your sentence, you need to break it into two slides.
- It is really important that your titles are visible. If someone daydreams when you are talking about the data on your slide, they will look to your titles in order to figure out what you were trying to say.
- Part of flow is consistency, so use the Slide Master tool in PowerPoint to set the size, location and font for your slide titles.

3.5. Conclusions/Future Directions Slide

- Words are allowed on this slide. It can even be a bulleted list.
- You must read this list out loud.
- For the purposes of this course and your thesis presentation, this could be a recap of your questions slide with the answers

3.6. Keep Slide Content Simple

- Each slide should convey one and only one idea
- An easy way to ensure this is to limit the number of figures per slide to one.

3.7. Convey your information with a visual picture/graph/scheme/cartoon.

- To lead your audience using *your* vision, you must show them what your vision is.
- This encourages the audience to start with your vision as a reference point. And leads to discussion about the merits of the vision as you see it.
- I will argue that you do not fully understand what you are doing or thinking if you cannot draw a picture of it.
- Creating visuals is a constant, ongoing process.

4. Suggestions for Effective Visuals

- 4.1. A picture says a thousand words, right? That is another reason why I have the “no words” rule.

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- It is much easier for people to convey your view with a picture
- Obviously, there are a few exceptions: sometimes you need words in order to convey your point (i.e. if you have a derivation).
- Except, do not show a derivation to a biological audience, or really almost any audience. Unless it is a Physics or Math department.
- If you do put more words (please don't), you should scrutinize whether the words are necessary.

4.2. Your slides must have a consistent color and cartoon scheme. This is part of the flow.

- Yes, this means you must draw your own pictures.
- And you must re-plot data from papers.
- You should be doing this anyway.
- If you ask my opinion on whether or not I think you should redraw a figure, I am going to reply yes about 99% of the time.

4.3. Slide Background Color Choices

- Use white backgrounds; do not use black backgrounds.
- White backgrounds project better in the vast majority of rooms. This is increasingly a challenge in the imaging field. I have not extensively played with this but there must be a better way to show fluorescent cells.

4.4. Graphical Object Color Choices

- Do not use green and red on the same slide. Ten percent of people are color blind and will not be able to see any distinctions between these colors. One trick you can use to check this is to print out your slides in grayscale.
- Do not use yellow or other really light colors. It is difficult to see data points that are yellow on a white background. If you have a white background (which you should), you should make sure that you can distinguish between the background and the data points.

4.5. Contrast Matters as Much as Color

- Contrast is the other graphical tool you can use to emphasize (or de-emphasize graphical items). Line widths for inserted shapes or figures should equal a minimum 2.25 pt for PowerPoint
- Dark colors draw the eye and are dominant. Therefore, use darker colors to draw the eye to what is most important on your slide, and use lighter colors to de-emphasize what is less important.
- The same trick you used to check your colors can be used to check contrast: print out your slides in grayscale and look at them.

4.6. Fonts and Font Size

- Do not use a distracting font. Use a simple font. I use Optima.
- Slide Titles should use font size 28. Size 24 is acceptable if it allows you to fit everything on one line. Do not go below size 20 font for titles. The default PowerPoint title font size is 44. That is too big.

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- Figures must have axis labels that are readable from the back of the room. Ideally you want the minimum font size to equal 18 or 24 in the final figure size. This is easy to do if you have only one figure per slide.
- Do not write full figure legends on slides.
- If you need to minimize the font so you can fit everything on there, you should reduce the content on the slide

4.7. Tables

- Avoid tables, if at all possible. Seriously. No Tables.
- Even a bar graph is better, but don't limit yourself to a bar graph if you can be more visually creative.
- But don't be too creative – you don't want your slides to be distracting.

5. Giving your Presentation

5.1. Interact with your slides.

- If you have good graphics, your slides will give you words. Just turn around and look at them and start describing, “The y-axis is fluorescence; the x-axis shows the ligand concentration. The data show a sigmoidal response to the ligand concentration...”
- Take your time to go through the slides.
- Only talk about what is on the slide; do not talk about anything that is not on the slide. If you are practicing and find yourself consistently adding “extra” information that is not on the slide, make another slide with that extra information.

5.2. Use professional language

- We scientistst obtain data or monitor a process, or measure a value. We do not *get* the data.
- Use scholarly language every time you practice. You will be nervous when you get up in front of a crowd for real and you will default to verbal habits, so develop the habit of using scholarly language when you practice with your peers and when you present at your lab meetings.
- On that note, practice your talk many times (especially the introduction). The introduction is where you will be most nervous. If you rock that, you will settle in faster and give a better talk.

5.3. Always use a microphone when one is offered.

- Do not try to get out of this by saying to the audience, “Can everyone hear me?”. No audience member wants to raise their hand to let everyone else that they are hard of hearing.
- Use the microphone skillfully. Practice with a microphone so you will know how to use it when the situation arises.
- Listen to yourself so that you can sense if the microphone is working effectively and/or if it is slipping and/or if its battery dies.
- It is exhausting to shout for 45 minutes.
- Place the microphone on your collar on the side that is closest to the slide so that you will speak into it more when you turn your head to look at the slide.

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5.4. Always use a pointer.

- Pointers are not very expensive, even the nice green ones that let you change slides. Purchase one so that you have your own pointer you are used to using.
- Practice with this pointer.
- Use it judiciously to interact with your slides in a calm, deliberate manner. Do not wildly move it around.
- Keep extra batteries in your bag. Prior to the talk, take out the batteries and place them on the podium so you won't have to fumble to find them if you need to change batteries in the middle of a talk.

5.5. Verbal Pace and Intonation

- Project but do not yell.
- Do not mumble. People want to hear what you have to say. As much as you might want to crawl under a rock, do not do it.
- Do not talk too fast. Most people talk faster when they are nervous and need to slow down. Very few people talk slower when they presenting.
- Avoid verbal fillers, like "um", "uh", etc. Instead, do not be afraid of silence. A slower pace with silent pauses is more effective than listening to someone stumble over their words as they try to talk faster than they can think.
- Do not speak in a monotone. Part of the mechanism for generating interest while you are telling your story is intonation in your voice pitch and word rate. Be enthusiastic about your work but not overly excited.

6. Question and Answer Period

6.1. Always repeat the question

- The goal with this rule is to be *reflective* not reactive.
- Also, most of the audience cannot hear the questioner asking the question.
- The only exception to this rule is when the questioner has a microphone themselves and is using it effectively.

6.2. Let the questioner completely finish their statement before you talk. You don't know what they are going to ask if you don't let them finish. When they are done, you should repeat/rephrase the question before you answer it.

- If you don't like the question, you can often redirect it to something you like.
- Sometimes the questioner is thinking out loud and has not fully formulated their question. Rephrasing the question allows you to restate it in a form you can answer.
- If members of the audience (like faculty) start having a discussion amongst themselves about your talk or about a question someone has asked, just let them talk. They will finish soon enough. There is no need for you to interrupt them. If you have something to say with added value, then share that when they finish their conversation. Otherwise, say thank you when they finish and look out at the audience to determine if there are more questions.